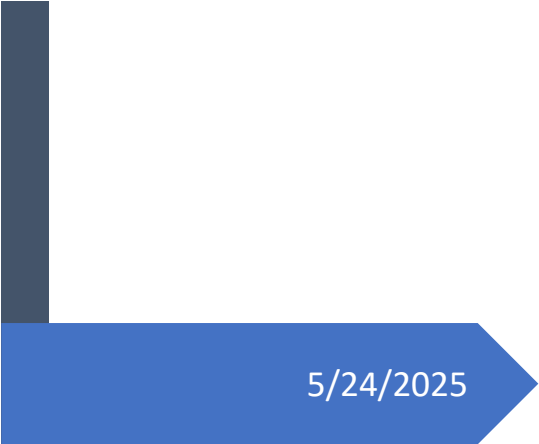
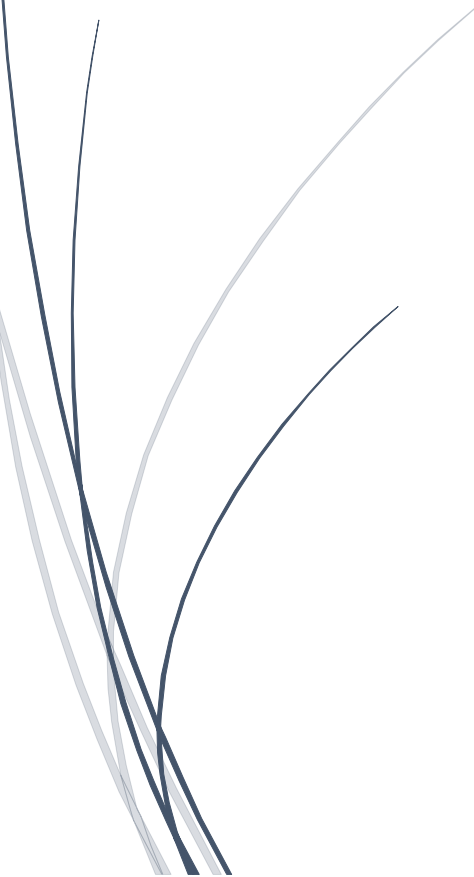


A dark blue vertical bar is on the left. A blue arrow points right from it, containing the date.

5/24/2025

T.O.A

Assignment#4

Several thin, curved lines in dark blue and light grey originate from the left side and curve upwards and to the right.

Mian Daniyal Hassan
SAP ID# 70150006

Question : $\{w \in \{0, 1\}^* \mid w \text{ does not start with } 01\}$

$R.E = (00+10+11)(1+0)^*$

$S \rightarrow AD$

$S \rightarrow BD$

$S \rightarrow CD$

$S \rightarrow \text{null}$

$S \rightarrow 0$

$S \rightarrow 1$

$D \rightarrow 1D$

$D \rightarrow 0D$

$D \rightarrow \text{null}$

$A \rightarrow 00$

$B \rightarrow 10$

$C \rightarrow 11$

Question : All words that start with a and have even length over alphabet set $\{a, b\}$.

$R.E = a(a+b)((a+b)(a+b))^*$

$S \rightarrow aA$

$S \rightarrow aAX$

$A \rightarrow a$

$A \rightarrow b$

$X \rightarrow AAX$

$X \rightarrow \text{null}$

Question : All words that start with b and have odd length over alphabet set {a, b},..

$R.E = b((a+b)(a+b))^*$

$S \rightarrow b$

$S \rightarrow bX$

$X \rightarrow AAX$

$X \rightarrow \text{null}$

$A \rightarrow a$

$A \rightarrow b$

Question : $\{ a^n b^m c^m d^{2n} \mid n \geq 0, m > 0 \}$.

$S \rightarrow AB$

$A \rightarrow aAdd$

$A \rightarrow \text{null}$

$B \rightarrow bBc$

$B \rightarrow bc$

Question : $\{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i + j = k\}$

$S \rightarrow aSc$

$S \rightarrow bSc$

$S \rightarrow \text{null}$

Picture's of rough work

CFG

① $\{0,1\}$ don't start with 01

$$W = \{00, 10, 11, \dots\}$$

$$R.E = (00 + 10 + 11)^* (1 + 0)^*$$

A B C D

$$\begin{aligned} S &\rightarrow AD = \cancel{S \rightarrow A} \rightarrow A & S &\rightarrow 1 \text{ valid words} \\ S &\rightarrow BD = \cancel{S \rightarrow B} \rightarrow B \\ S &\rightarrow CD = \cancel{S \rightarrow C} \rightarrow C \end{aligned}$$

$$\begin{aligned} D &\rightarrow 1D \\ D &\rightarrow 0D \rightarrow \text{loop} \\ D &\rightarrow 1 \end{aligned} \quad \begin{aligned} S &\rightarrow 0 \\ S &\rightarrow 1 \end{aligned}$$

$$\begin{aligned} A &\rightarrow 00 \\ B &\rightarrow 10 \\ C &\rightarrow 11 \end{aligned}$$

② start with a & have even length

$$\frac{a}{\cancel{a}} (\frac{(a+b)(a+b)(a+b)}{X})^* + \frac{a(a+b)}{B}$$

$$\begin{aligned} S &\rightarrow aB \rightarrow \text{valid words} \\ \cancel{S \rightarrow aX} \\ S &\rightarrow aX \end{aligned}$$

$$\begin{aligned} X &\rightarrow BXB \rightarrow X \text{ is value} \\ B &\rightarrow aB \rightarrow B \text{ is value} \\ B &\rightarrow b \end{aligned}$$

$$(3) \quad R.E = \underline{b(a+b)(a+b)}^A$$

X

$$\begin{aligned} S &\rightarrow b \\ S &\rightarrow bX \\ X &\rightarrow AX \mid \Lambda \\ A &\rightarrow a \\ A &\rightarrow b \end{aligned}$$

$$(5) \quad \{ \underbrace{a^n}_A \underbrace{b^m c^m}_B \underbrace{d^{2n}} \mid n \geq 0, m \geq 0 \}$$

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow aA \mid d \\ A &\rightarrow \Lambda \\ B &\rightarrow bc \\ B &\rightarrow bBc \end{aligned}$$

$$(5) \quad \{ a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i+j=k \}$$

$$\begin{aligned} S &\rightarrow \cancel{A} S B c \\ A &\rightarrow a A \mid \Lambda \\ B &\rightarrow b B \mid \Lambda \\ C &\rightarrow c C \mid \Lambda \end{aligned}$$

$$\begin{aligned} S &\rightarrow a S c \\ S &\rightarrow b S c \\ S &\rightarrow \text{null} \end{aligned}$$

(2)

$$R.E = a(\underbrace{a+l}_A) + a\underbrace{a+l}_{\cancel{A}}(\underbrace{a+l}_{\cancel{A}})(\underbrace{a+l}_{\cancel{A}})^*$$

~~$S \rightarrow a$~~
 ~~$S \rightarrow aA$~~
 ~~$S \rightarrow aX$~~
 ~~$A \rightarrow a$~~
 ~~$A \rightarrow l$~~
 ~~$X \rightarrow AAAX$~~
 ~~$X \rightarrow \Lambda$~~

$$R.E = \underbrace{a}_{\cancel{A}}(\underbrace{a+l}_A)(\underbrace{a+l}_{\cancel{A}})(\underbrace{a+l}_{\cancel{A}})^*$$

$S \rightarrow aA$
 $S \rightarrow aAX$
 $A \rightarrow a$
 $A \rightarrow l$
 $X \rightarrow AAAX$
 $X \rightarrow \Lambda$